

Global Population Estimates & Projections: Urbanisation Trends (1960–2050)

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1. Introduction

Urbanisation is one of the most transformative demographic forces shaping the modern world. I chose to explore it through a combination of statistical reasoning, Power BI techniques, and structured data storytelling. As I began working with the datasets provided, I aimed not only to create a visually appealing dashboard but also to truly understand how the world's population has shifted over the past decades—and how it is expected to continue shifting through 2050.

The process of developing this dashboard felt like a journey. I began by exploring the raw datasets, identifying the most pertinent indicators, and thoughtfully considering how best to communicate complex global patterns to users who may lack a statistical background. From the outset, I aimed to make the dashboard intuitive, interactive, and meaningful, reflecting both academic best practices and my own understanding of data design.

2. Dataset

The datasets used in this project were supplied by the University as part of the ASDV module resources. I used the three datasets provided in the coursework instructions. Together, they gave me a complete picture of how different countries and regions have developed from 1960 to projected estimates in 2050.

- Population Figures & Projections (1960–2050)

This dataset served as the primary foundation of my analysis. It included values for urban population, rural population, male and female totals, growth rates, and life expectancy. The dataset also contained projected values beyond 2022.

- Metadata File

Before analysing any field, I reviewed the metadata to grasp the definitions and limitations. This process helped me confidently interpret what each indicator signified.

- Country Groupings

This dataset allowed me to group countries into regions—such as South Asia, Sub-Saharan Africa, and Europe & Central Asia—making it easier to compare urbanisation patterns across the world.

Working with these datasets helped me deeply understand how demographic information is collected and structured. It also guided me in selecting the appropriate variables for the dashboard I eventually built in Power BI Desktop.

3. Data Preparation

I did all data preparation for this project using Python and the pandas library. Since Power BI Web does not include a Power Query interface for data transformation, using Python provided a reliable, transparent environment for cleaning, reshaping, and merging datasets before importing them into Power BI for interactive dashboard development.

The cleaning workflow was separated into clear steps to ensure reproducibility and to allow each transformation to be traced back to a well-defined part of the process.

3.1 Loading the Raw Files

The first stage involved loading the two source datasets:

- Population Figures & Projections.csv
- Country Groupings.xlsx

These files were imported using `pd.read_csv()` and `pd.read_excel()`.

```
import pandas as pd

# Load population dataset
pop_file = "Population Figures & Projections.csv"
df_population = pd.read_csv(pop_file)

# Load country groupings dataset
group_file = "Country Groupings.xlsx"
df_country_groupings = pd.read_excel(group_file, sheet_name="List of economies")

df_population.head(), df_country_groupings.head()
```

This step allowed me to inspect their structure and confirm the presence of key indicators, such as urban and rural populations, and the country codes required to join the files.

3.2 Selecting and Renaming Columns

The population dataset contained many fields that were not required for this project.

I selected only the relevant variables:

- Country Code
- Country Name
- Year
- Urban Population
- Rural Population

These were then renamed to shorter, clearer column names.

```
# Select required columns and rename them
pop_cols_map = {
    'Country Code': 'Code',
    'Country Name': 'Country',
    'Time': 'Year',
    'Rural population [SP.RUR.TOTL]': 'Rural Population',
    'Urban population [SP.URB.TOTL]': 'Urban Population'
}

df_pop = df_population[list(pop_cols_map.keys())].rename(columns=pop_cols_map)
df_pop.head()
```

This step streamlined the dataset and made later transformations easier to understand.

3.3 Converting Population Columns to Numeric

Several population fields were stored as text.

Using `pd.to_numeric()` ensured that:

- Urban Population
- Rural Population

were properly recognised as numeric fields.

```
# Convert to numeric, coerce non-numeric values to NaN
df_pop['Rural Population'] = pd.to_numeric(df_pop['Rural Population'], errors='coerce')
df_pop['Urban Population'] = pd.to_numeric(df_pop['Urban Population'], errors='coerce')

df_pop.info()
```

This prevented errors during mathematical operations and made the dataset consistent.

3.4 Creating Total Population

Since the dataset did not include a direct total population variable, I created one:

```
# Creating Total Population
df_pop['Total Population'] = df_pop['Urban Population'] + df_pop['Rural Population']

df_pop[['Urban Population', 'Rural Population', 'Total Population']].head()
```

This new field makes it clearer for comparisons and is also needed for calculating the global urbanisation rate.

3.5 Cleaning and Preparing Country Groupings

The country groupings dataset was filtered to include only:

- Code
- Region
- Income Group

Column names were harmonised to ensure compatibility during merging.

```
# Select and rename grouping columns
group_cols = ['Code', 'Region', 'Income group']
df_group = df_country_groupings[group_cols].rename(columns={'Income group': 'Income Group'})
df_group.head()
```

This dataset adds essential regional context to the population data.

3.6 Merging Both Datasets

Using a left join on Country Code, I merged:

- the cleaned population dataset
- with
- the cleaned country groupings table

```
# Left join population with groupings
df_merged = pd.merge(df_pop, df_group, on='Code', how='left')

# Replace missing grouping values
df_merged['Region'] = df_merged['Region'].fillna('Aggregates')
df_merged['Income Group'] = df_merged['Income Group'].fillna('Aggregates')

df_merged.head()
```

Missing region and income-group values were labelled as "Aggregates" to keep the dataset complete. This merging step produced a unified table containing population metrics and region classifications.

3.7 Calculating Urbanisation Rate

Urbanisation Rate was calculated as:

```
# Calculate Urbanisation Rate
df_merged['Urbanization Rate'] = (
    df_merged['Urban Population'] / df_merged['Total Population']
).fillna(0)

df_merged[['Urban Population', 'Total Population', 'Urbanization Rate']].head()
```

This measure is central to understanding long-term demographic change and has become one of the key variables used across the dashboard.

3.8 Filtering the Analysis Years

The project specification requires analysis between 1960 and 2050.

```
# Filter dataset to required years
df_final = df_merged[(df_merged['Year'] >= 1960) & (df_merged['Year'] <= 2050)]

df_final.head()
```

The dataset was filtered to remove irrelevant years while preserving the full historical and projected timespan needed for the dashboard.

3.9 Exporting the Final Clean Dataset

I then exported the fully prepared dataset :

```
output_filename = "master_population_data_clean.csv"
df_final.to_csv(output_filename, index=False)

print( output_filename)

master_population_data_clean.csv
```

This file was then uploaded into Power BI Web, where I built all visualisations, KPIs, slicers, and interactive elements.

3.10 Final Verification

A random sample of rows was inspected to ensure:

- Merging was successful
- Population values were correct
- No unexpected missing values
- All years were included
- The Urbanisation Rate was calculated properly

df_final.sample(10)									
	Code	Country	Year	Rural Population	Urban Population	Total Population	Region	Income Group	Urbanization Rate
1541	BRB	Barbados	2045	168016.0	105844.0	273860.0	Latin America & Caribbean	High income	0.386489
3726	CHL	Chile	2046	1807358.0	18881267.0	20688625.0	Latin America & Caribbean	High income	0.912640
9511	KOR	Korea, Rep.	2007	8942697.0	39740941.0	48683638.0	East Asia & Pacific	High income	0.816310
12861	NZL	New Zealand	1990	508061.0	2821739.0	3329800.0	East Asia & Pacific	High income	0.847420
17488	THA	Thailand	1976	31677548.0	10204580.0	41882128.0	East Asia & Pacific	Upper middle income	0.243650
17229	SYR	Syrian Arab Republic	1990	6337150.0	6071846.0	12408996.0	Middle East & North Africa	Low income	0.489310
14663	RUS	Russian Federation	1972	47384351.0	84524649.0	131909000.0	Europe & Central Asia	High income	0.640780
15123	SAU	Saudi Arabia	1977	3377676.0	5377543.0	8755219.0	Middle East & North Africa	High income	0.614210
1725	BEL	Belgium	2047	142371.0	12005320.0	12147691.0	Europe & Central Asia	High income	0.988280
9072	JOR	Jordan	2023	904697.0	10432355.0	11337052.0	Middle East & North Africa	Lower middle income	0.920200

This confirmed that the dataset was ready for analysis.

4. Dashboard Design

The dashboard was designed as a single-screen, highly interactive interface to support rapid insight into global urbanisation patterns between 1960 and 2050. The dashboard emphasises clarity, simplicity, and analytical depth while maintaining an intuitive user experience.

4.1 Overall Layout and Design Principles

The dashboard follows a top-down analytical hierarchy:

- Top row: KPIs
- Middle section: Main trend visual
- Bottom section: Regional and detailed breakdowns

4.2 KPI Cards

The first row presents four KPIs:

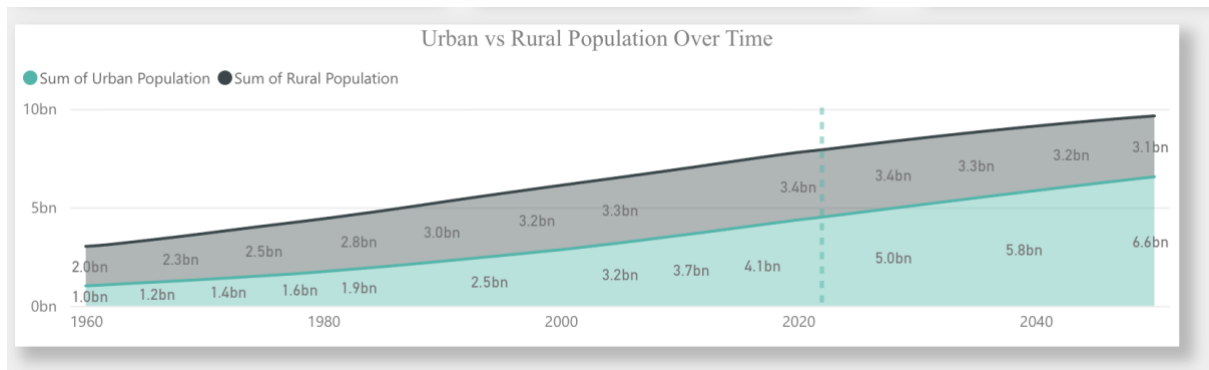
- 1960 Global Population
- 2022 Global Population
- 2050 Projected Population
- Average Urbanisation Rate (2022)



Formatting decisions:

- Rounded values shown in billions to support cognitive load reduction (Tufte, 2001).
- Titles were manually renamed and category labels removed to improve clarity.
- Decimal precision set to two places for accuracy and uniformity.

4.3 Main Visual: Urban vs Rural Population Over Time

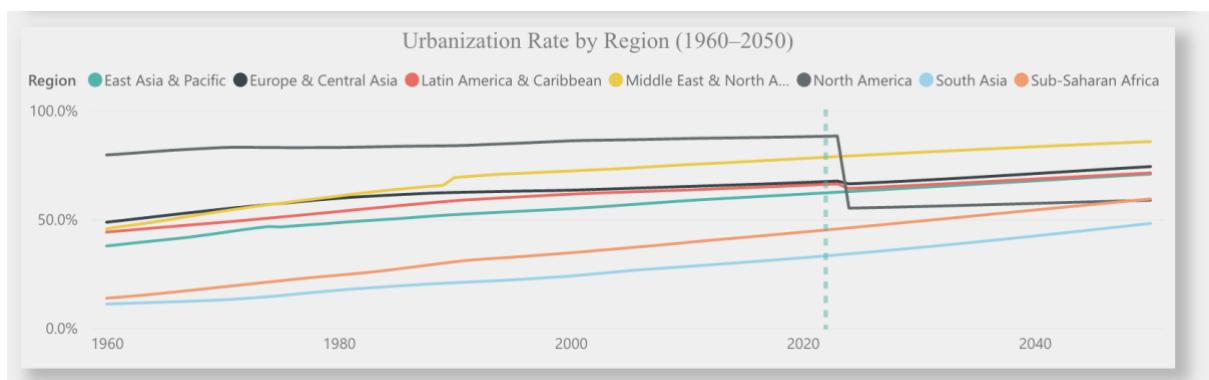


The central visual is a stacked filled line chart, chosen because:

- It clearly communicates long-term change
- It reveals the crossover point where the urban population surpasses the rural population (~2007–2010)
- It emphasises the magnitude and the relationship between the two measures

Annotations such as data labels, a vertical visual marker, and colour separation enhance interpretability.

4.4 Regional Urbanisation Trends (1960–2050)



A multi-line chart was used to compare urbanisation rates across global regions.

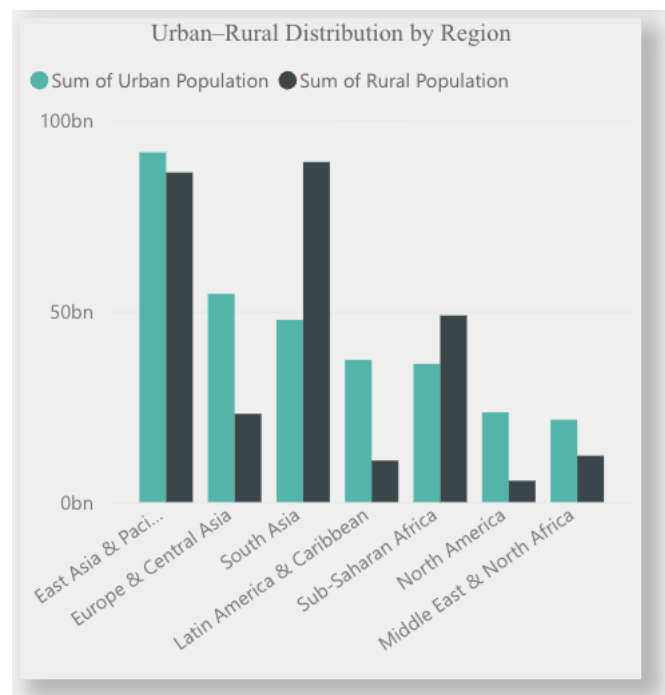
This design was chosen because:

- It supports multi-series comparison
- It allows immediate recognition of regional differences in pace and level of urbanisation
- Colour encoding maintains consistent mapping across visuals

This chart highlights that North America, Europe and Latin America were early urbanisers, while South Asia and Sub-Saharan Africa show slower but accelerating growth.

4.5 Regional Urban–Rural Distribution Bar Chart

I used a clustered bar chart to compare the distribution of urban and rural populations across global regions. Unlike the line charts and KPI cards, this visual responds to the Year slicer, allowing users to examine regional population structures for any chosen year, whether historical (e.g., 1960) or projected (e.g., 2050).



Justification:

- Bar charts are ideal for comparing categories such as world regions
- Vertical axis supports clearer reading of magnitude differences
- Colours match the palette used in earlier visuals for visual consistency
- The selective interaction design (slicer affects only this chart) enables users to explore yearly regional differences without altering the broader global trends shown in the line charts

This approach demonstrates controlled interactivity, ensuring clarity while still providing user-driven exploration where it is most meaningful.

5. Analysis & Key Findings

This presents the key insights derived from the dashboard and highlights major global and regional population trends between 1960 and 2050, with the combination of KPIs, time-series charts, and region-based comparisons providing a structured analytical narrative that reveals both long-term global patterns and region-specific dynamics.

5.1 Global Population Growth (1960–2050)

The global population has increased dramatically over the past decades. According to the dashboard's KPI indicators, the world population grew from approximately 3 billion in 1960 to almost 8 billion in 2022, representing more than a doubling within a single lifetime. The Projections indicated a further rise to around 9.65 billion by 2050, although the growth rate began to slow down after 2030.

This upward trend is clearly visible in the time-series line charts. The Total Population curve demonstrates a smooth, continuous rise, reflecting the improvements in healthcare, declining infant mortality, and increasing life expectancy.

5.2 Urban vs Rural Population Dynamics

One of the most significant global transformations shown in the dashboard is the shift from a mainly rural world to a mainly urban one. In 1960, rural populations exceeded urban populations by a substantial margin. However, rapid urban growth—shown by the steep ascent of the urban population line—led to a historic crossover around 2007–2010, when the world became majority-urban for the first time.

By 2050, urban populations are projected to reach roughly 6.6 billion, while rural populations are expected to decline to around 3.1 billion. This shift has profound implications for housing, employment, transport, and infrastructure worldwide.

5.3 Global Urbanisation Rate Trends

The Average Urbanisation Rate KPI reveals that the world was around 61% urban in 2022—a dramatic increase from around 33% in 1960. Projections in the dashboard suggested a global urbanisation rate of nearly 70% by 2050, indicating that cities will continue to dominate economic and demographic landscapes.

The time-series chart for regions shows that:

- North America and Latin America have been highly urbanised for decades.
- Europe & Central Asia show stable, gradual increases.
- East Asia & Pacific experienced rapid urbanisation due to industrialisation and economic reforms.
- South Asia and Sub-Saharan Africa started from very low bases but show steep upward trajectories, signalling accelerating urban transitions.

5.4 Regional Urbanisation Patterns

The regional line chart enables comparisons across major world regions, revealing significant differences:

- North America leads with urbanisation rates consistently above 80%.
- Latin America & the Caribbean urbanised very early and remains one of the most urbanised regions globally.
- South Asia and Sub-Saharan Africa remain less urbanised overall but demonstrate the fastest growth rates.
- The Middle East & North Africa follow a middle-growth trajectory influenced by economic diversification and urban development policies.

This highlights that urbanisation is not a uniform global process but varies considerably by region.

5.5 Regional Urban–Rural Structures for Selected Years (Bar Chart)

The clustered bar chart, which updates based on the Year slicer, reveals how each region's population structure changes over time.

Key insights include:

- East Asia & Pacific has the highest population overall, with urban populations increasing markedly after 2000.
- Sub-Saharan Africa shows a substantial increase in both urban and rural populations, but urban growth is faster, indicating a region undergoing its most transformative urbanisation phase.
- North America and Europe & Central Asia show little rural population growth, reflecting mature urban systems.
- South Asia transitions from mainly rural to more urbanised as fertility declines and major cities expand.

This underscores the importance of region-specific strategies for managing demographic changes.

6. Discussion and Conclusion

The dashboard provides a clear, interactive view of global population and urbanisation trends between 1960 and 2050. The visuals show steady global population growth, the shift from a mainly rural world to a mainly urban one, and the differences in urbanisation patterns across regions. The use of controlled interactivity—where the Year slicer affects only the regional bar chart—helps users explore specific years without altering the global trendlines or KPIs. The design choices, such as consistent colours and a clean layout, make the dashboard easy to interpret and aligned with best practice.

Although the dashboard is limited to the variables included in the dataset, it effectively communicates both historical and projected population trends. Overall, the project transforms raw data into a clear and informative tool that helps users understand long-term demographic change.

7. Reference

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